Ravenna – network neologism

The founding of the first ancient lagoon settlement of Ravenna is difficult to attribute to one tribe but it prospered in the Roman Empire and survived all the vicissitudes after its fall. Today, Ravenna is twinned with six towns across Europe, including ROB JAMES' local town of Chichester, and so perhaps it's an apt name for a new open network audio standard that must encompass the many and various requirements of some very different bedfellows.

Announced formally with some considerable bally-hoo at the 2010 IBC in Amsterdam by ALC NetworX, whose brainchild it is under the direction of CEO Philipp Lawo, the Ravenna consortium already includes a number of partners – Genelec, Lawo, Merging Technologies, Innovason, DirectOut, LSB/VSM Control, DSA Volgmann, and Schoeps.

So, what is Ravenna and why is it relevant to audio professionals? In short, it represents a new take on the third generation form of audio interconnect, where the first generation of interconnect is analogue point-to point copper, the second generation uses digital codes representing the analogue signal, conveyed point to point over copper or fibre-optic cabling, and the third generation also employs digital codes representing the analogue audio but transported as packets over network infrastructure.

Conventional real-time digital audio (and video) switches/routers are highly specialised and expensive devices designed and built specifically to distribute large numbers of signals with minimal delay with bit-for-bit veracity. Interest in Audio-over-IP is burgeoning among broadcasters and others principally for reasons of cost and flexibility. Network based audio and video distribution systems rely on off-the-shelf IT technology designed originally to forward large numbers of data packets with no constraints on real-time delivery. For this reason standard IP technology does have some disadvantages for audio transport. However, the massive scale of data networking equipment production means lower costs and the opportunity to leverage existing infrastructure is too seductive to ignore, especially since it offers the promise of a scaleable platform for future revenue earning developments such as HD radio and data services.

Ravenna is very well suited to areas where complex audio routing/mixing systems are deployed. For example, in-house distribution in broadcasting centres and WAN connections to satellite studios, OB vans, where hook up to venues with the same infrastructure becomes simple, in venues themselves for local signal distribution and connection to just such OB vehicles when required. For live events and concerts it offers highly flexible temporary installation possibilities and in theatres, opera houses and houses of worship it can provide low-cost local signal distribution. Notwithstanding all of the above, Ravenna is also an excellent candidate for relatively simple point to point interconnects such as computer to audio interface.

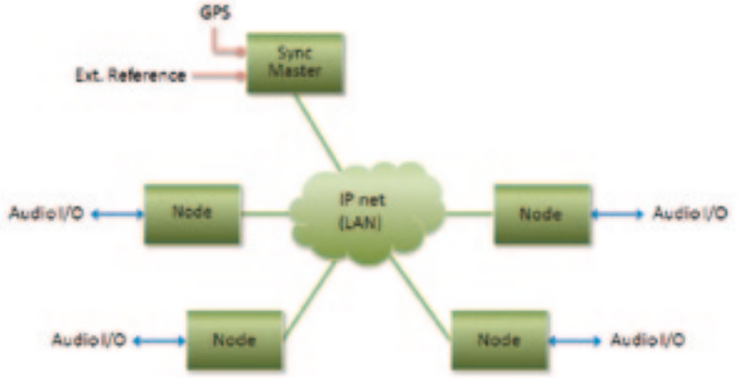
For years the most ingenious minds in the audio business have applied themselves to developing innovative connectivity solutions and there are already a number of proprietary audio solutions that use networking hardware. Sadly most of this endeavour has ended up being used in proprietary products or in a manufacturer's licensing portfolio. Some of these developments have been established as de facto standards and some, like MADI, for example, have even managed to achieve standards body status but mostly only by making the technology freely available to everyone.

However, Ravenna, leaving aside the other advantages touted, is an open standard based on the ubiquitous IP protocol. Specifically, protocol levels on or above layer 3 of the OSI reference model. Since Ravenna is purely based on layer 3 protocols, it can operate in most existing network environments. Unlike layer 1 or layer 2 solutions, it does not, in principle, require its own network infrastructure. IP can therefore be transported on virtually any LAN and is used as the base layer for communication across WAN connections (including the internet). Although in most cases Ethernet will be deployed as the underlying data link layer, IP is in general infrastructure-agnostic and can be used on virtually any network technology and topology. All the protocols and mechanisms used in Ravenna are based on well established and commonly used methods from the IT and audio industries and comply with various standards defined and maintained by the international standards bodies.

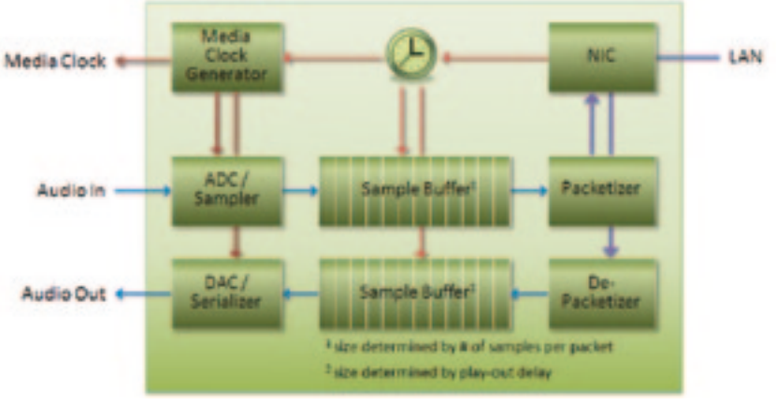
From a practical standpoint, Ravenna offers the tantalising prospect of real-time distribution of audio and other media content in IP-based network environments with simple interoperability between equipment sourced from a variety of manufacturers. Although aimed principally at the broadcast market Ravenna has obvious applications in recording, live sound and the installation markets.

The challenges with any audio transport based on IT are latency and accurate synchronism. Ravenna is designed from the ground up to meet the strict requirements of the pro audio market with low latency, full signal transparency and a high level of reliability. It employs several ingenious strategies to these ends and offers flexible configuration options to match performance to the specific requirements. While it is perfectly possible to use Ravenna over most existing network infrastructures for non time critical applications, such as journalist and news feeds, where sub-millisecond latency and absolute sync are the priority careful choice of network topology and using single sample per channel packets offers the lowest possible latency. Adding samples/channels up to the maximum allowed packet size guarantees the maximum number of channels throughput. A variety of stream formats with different sample rates can be used concurrently within the same network although format and sample rate selection may be limited by the capabilities of specific nodes.

A Ravenna system requires a carefully configured IP network, a master clock device and any number of Ravenna enabled I-O nodes. The master clock can be either a dedicated device or any Ravenna node capable of serving as a grandmaster. The preferred time domain reference is GPS. Simple streaming across a network can be achieved without any synchronisation at all but in pro audio applications tight synchronisation between all devices and streams is absolutely mandatory. While playback synchronisation in most applications requires sample accuracy, one goal for Ravenna is to provide superior performance by offering phase-accurate synchronisation as an option thus rendering separate reference Word clock distribution throughout a facility or venue redundant.



Ravenna overview block diagram showing all the basic components required for a Ravenna network: an IP network, Ravenna I-O nodes and a time grandmaster.



Functional block diagram of a Ravenna node.

A Ravenna node consists of a NIC (Network Interface Controller), a Media Clock which can slave to the house clock or be a master in its own right, A-D and D-A convertors and a packetiser and de-packetiser each with its own sample buffer. A single node can support many audio I-O channels depending on the design. Nodes can be configured remotely across the network and concurrent unicast and multicast operation is supported.

Unicast (one-to-one) is used in application scenarios such as an individual stream between two devices (e.g. a multichannel stream between a console and a recorder/DAW). This uses a point-to-point connection between the sender and

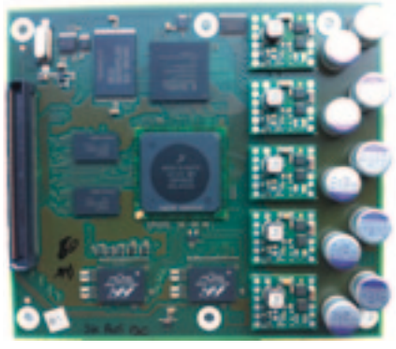
receiver. Since each additional receiver adds its own individual connection, network traffic increases with every additional unicast stream. Multicast (one-to-many) streaming is used in scenarios where a single source is to be distributed to many potential recipients (e.g. programme stream to journalists' desktops). At the sending end this only requires one connection per stream. Network switches are aware of which participants (receivers) should receive any particular multicast and forward packets only to registered nodes. In multicast set-ups the network traffic only increases on the last (closest to receiver node) segment(s) of the network path.

The network infrastructure must be able to transport IP packets and must support a number of standard operating protocols, e.g. RTP/RTPC for streaming since this is used widely and supports a wide variety of standard payload formats. Some of these formats will be mandatory for all Ravenna devices, others will be optional. For example, this protocol offers the possibility of standard media player applications subscribing to Ravenna streams. Synchronisation across all nodes is achieved via IEEE1588-2008 (PTPv2 Precision Time Protocol). This is another standard protocol that can be used on IP. PTPv2 provides a means for synchronising local clocks to a precision defined in AES-11. Accurate synchronisation can even be achieved across WAN connections when GPS is used as a common time domain.

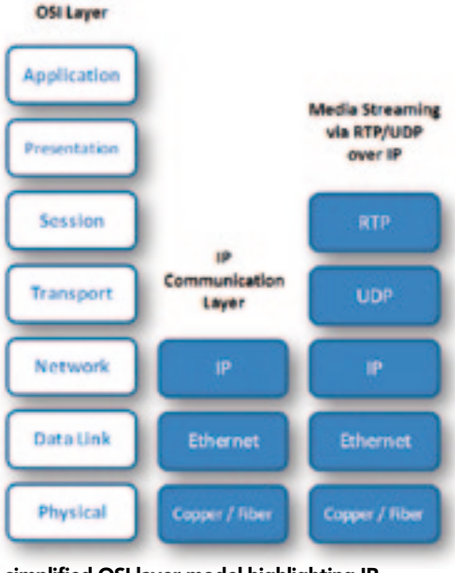
For the QoS (Quality of Service) protocol DiffServ has been chosen since it is widely supported by most modern managed switches. Since other traffic can coexist with Ravenna on the same network, Ravenna traffic must be on the fast track. Ravenna packets are assigned a high priority classification to ensure expedited transport across the network, while other packets with lower priority are treated as best-effort traffic. Even within Ravenna there are different priorities assigned to different classes of traffic. Synchronisation is assigned the highest priority, immediately followed by any real-time media traffic, while control and configuration traffic will be on a lower priority level. Any non-Ravenna traffic would receive the lowest (standard) priority and be treated as best-effort traffic. Performance and capacity scale with the capabilities of the underlying network architecture.

As you would expect for a system aimed at broadcast, Ravenna supports redundancy. Although modern network infrastructures can be configured to guarantee a high level of transport security and reliable 24/7 operation, for added security there is the option of full network redundancy. Each Ravenna device can include two independent network interfaces, which can be connected to independent physical networks. By duplicating any outgoing stream to both network links, any destination device will receive the full stream data on both network interfaces independently. If data from one link is corrupted, or one network link fails completely, the uncorrupted data is still present on the other link. Changeover in the event of the failure of a network link is automatic.

The system design approach allows for operation with or without centralised services for configuration/connection management. ALC NetworX recommends that basic device configuration (e.g. initial settings and setup of audio streams) should be executed via a web interface (http). However, other methods may be used in addition or as an alternative. Device discovery is accomplished with DNS-SD (via an mDNS or DNS service). In small networks, without DHCP/DNS servers, the zeroconf mechanism — a fully automatic, self-configuring method — is used for auto-IP assignment and service advertisement and discovery. Streams available on the network are represented by SDP records with extended information (i.e. a clock domain identifier, RTP time stamp association, etc.) Clients can connect to streams via RTSP or SDP/http.



Com8313 piggyback module used in some of the technology prototypes shown at IBC (Innovason's Eclipse, Direct Out's Producer. Com/R, Lawo's Dallis and Crystal).



A simplified OSI layer model highlighting IP protocol layer and RTP streaming layer.

Ravenna is arguably the most promising audio industry interconnect technology we have seen for a long time since it offers low cost of implementation to end users and equipment manufacturers while being able not only to replace existing expensive and proprietary interface technologies but also to expand capability. Apart from the obvious utility of being able to send large numbers of channels between different manufacturers kit, reliably, with low latency where required and impeccable reproduction, Ravenna's IP nature makes it a natural for remote control of devices as diverse as digital microphones and loudspeakers as well as the more obvious consoles, DAWs and recorders.

When implemented fully at enterprise level Ravenna can satisfy a huge range of needs from office loop feeds to pretty much every conceivable interconnect application in production and recording.

Most importantly, unlike most other existing networking solutions, Ravenna is an open technology standard without a proprietary licensing policy. ALC NetworX is actively encouraging any interested parties to participate in the enhancement and evolution of its capabilities. ALC NetworX is also happy to assist manufacturers interested in implementing Ravenna into their products, since broad market acceptance will depend on take-up. As a result of this enlightened attitude and Ravenna's many virtues I confidently expect to see many other manufacturers embrace the Ravenna philosophy in the next months and years. ■